

Mark schemes

Q1.

(a) (i) $P(SI) = 0.3$

B1
1

(ii) $P(SI \& A) = 0.3 \times 0.2 = 0.06$
ft wrong answer to (i)

B1F
1

(iii) $P(SI' \& A) = 0.7 \times 0.4$
M1 for reasonable attempt

M1

$P(A) = 0.06 + 0.28 = 0.34$
convincingly shown

A1
2

(b) Cond prob with N or D correct
where $0 < N < D < 1$

M1

Numerator 0.06
ft wrong answer to (a)(ii)

A1F

Denominator 0.34

A1
3

[7]

Q2.

(a) (i) $P(\text{both}) = 0.1 \times 0.05 = 0.005$

M1A
2

(ii) $P(\text{neither}) = 0.9 \times 0.95 = 0.855$

M1A1
2

- (iii) $P(\text{exactly one}) = 0.14$
ft wrong values if subtraction from 1 used

M1A1F

2

- (b) Formula for conditional prob
*Fraction with $0 < N < D < 1$ and D correct
 or equal to c 's answer to (a)(iii)*

M1

Numerator = 0.1×0.95

m1

Denom = 0.14 so ans = $\frac{19}{28}$
*ft wrong answer to (a)(iii);
 Accept AWR 0.679 or 0.678*

A1F

3

[9]

Q3.

- (a) (i) $P(\text{British}) = \frac{13}{30}$
Allow decimal answers throughout question

B1

1

(ii) $P(\text{goalkeeper}) = \frac{4}{30} \left(\frac{2}{15} \right)$

B1

1

(iii) $P(\text{British \& goalkeeper}) = \frac{2}{30} \left(= \frac{1}{15} \right)$

B1

1

(iv) $P(\text{defender or midfielder}) = \frac{21}{30} \left(= \frac{7}{10} \right)$

B1

1

(v) $P(\text{Br or attacker, not both}) = \frac{14}{30} \left(= \frac{7}{15} \right)$
*M1 for **fully** correct method*

M1A1

2

(b) (i) $P(\text{goalkeeper} | \text{British}) = \frac{2}{13}$
M1 for correct denominator

M1A1

2

(ii) Not independent (reason)
E1 if incompletely explained eg no numerical value

E2,1

2

[10]

Q4.

(a) (i) $P(\text{work \& TMR}) = \frac{2}{3} \times \frac{2}{5} = \frac{4}{15}$
'TMR' means 'ten mile run'

M1A1

2

(ii) $P(\text{no work, TMR}) = \frac{1}{3} \times \frac{4}{5} = \frac{4}{15}$

M1A1

2

(iii) $P(\text{TMR}) = \frac{4}{15} + \frac{4}{15} = \frac{8}{15}$
ft errors in (i) and /or (ii)

A1F

(b) Reasonable attempt

$\frac{x}{y}$
 Allow $\frac{x}{y}$ where $0 < x < y = (a)(iii)$

M1

$P(\text{work} | \text{TMR}) = \frac{1}{2}$

Allow NMS; ft errors in (a)

A1F

2

[7]

Q5.

(a) (i) $P(\text{treat}) = 0.6 \times 0.8 = 0.48$

M1 for attempt to multiply probs appropriately

M1A1

2

(ii) Use of $P(\text{behaviour not good}) = 0.4$

M1A0 for MR:

$P(\text{good beh, no treat}) = 0.12$

M1

$P(\text{HW ex't, no treat}) = 0.4 \times 0.5 = 0.2$

A1

2

(iii) $P(\text{HW excellent}) = 0.48 + 0.2 = 0.68$

ft wrong answer to (i) and/or (ii)

B1F

1

(b) Use of formula for conditional prob

with correct (ft) N or D, $0 < N < D < 1$

M1

Numerator 0.2

ft wrong answer to (a)(ii)

A1F

Denom 0.68 (so ans is $\frac{5}{17} \approx 0.29$)

ft wrong answer to (a)(iii)

A1F

3

[8]

Q6.

(a) (i) $P(\text{both}) = 0.9 \times 0.3 = 0.27$

M1A1 2

(ii) $P(\text{neither}) = 0.1 \times 0.7 = 0.07$

M1A1 2

(iii) $P(\text{at least one}) = 1 - 0.07 = 0.93$
ft wrong answer to (ii)

M1A1F 2

(b) Conditional probability = $\frac{0.9}{0.93} \left(= \frac{90}{93} \right)$
ft wrong answer to (a)(iii) provided it lies between 0.9 and 1
Condone failure to simplify $\frac{0.9}{0.93}$
NMS 2/3 for AWRT 0.97

B3F 3

[9]

Q7.

(a) $P(N \cap M) = 0.8 \times 0.9 (= 0.72)$
S.R. If $\frac{0}{5}$ then B1 for accurate tree diagram.

$P(\bar{N} \cap M) = 0.2 \times 0.05 (= 0.01)$

A1

$P(M) = 0.72 + 0.01 = 0.73$
first M1 applies to either answer

M1A1 5

(b) $P(N/M) = \frac{P(N \cap M)}{P(M)} = \frac{0.72}{0.73}$
top line f.t.
bottom line f.t.

M1/M1 3

= 98.6% AEF

e.g. $\frac{72}{73}$

A1

[8]

Q8.

- (a) 0.7×0.8 and/or 0.3×0.4 seen

*May appear on copy of tree diagram;
allow M1 even if candidate's answer is 0.56 or 0.12*

M1

Req'd prob = $(0.7 \times 0.8) + (0.3 \times 0.4)$

m1

... = 0.68

A1

3

(b) $P(\text{in | wins}) = \frac{0.56}{x}$
 $0 < x < 1$

M1

$x = 0.68$

ft wrong answer to (a)

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A1F

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Answer $\frac{14}{17}$

OE; accept AWRT 0.82

NMS 3/3

A1

3

[6]

Q9.

- (a) (i) $P(\text{female \& Sp}) = 0.2$

AG; M1 for appropriate multiplication

M1A1

2

- (ii) $P(\text{male} \& \text{Sp}) = 0.1$
OE; the M1 can be earned here

A1

2

So required probability is 0.3
ft one error

A1F

- (b) Formula for conditional probability
With correct (ft) N or D , $0 < N < D < 1$

M1

3

Numerator 0.2
ft if c's answer to (a)(i) used here

A1F

Denominator 0.3
ft wrong answer to (a)(ii)

A1F

[7]

Q10.

- (a) (i) $P(\text{female}) = 45/75 = 0.6$

M1

- (ii) $P(\text{female} \mid >8) = 13/20 = 0.65$

M1

- (iii) $P(\text{female} \mid <3) = 12/20 = 0.6$

M1

acf

A1

4

- (b) Independent $P(\text{female}) = P(\text{female} \mid <3)$

M1

*comparison of $P(\text{female})$ with
 $P(\text{female} \mid <3)$
 or $P(<3)$ with $P(<3 \mid \text{female})$
 or $P(\text{female}) \times P(<3 \mid \text{female})$ with $P(\text{female} \& <3)$
 or $P(<3) \times P(\text{female} \mid <3)$ with $P(\text{female} \& <3)$*

correct conclusion from correct probabilities

A1

2

- (c) selecting a male teacher
any mutually exclusive event

B2,1

2

- (d) (i) $\frac{12}{75} \times \frac{11}{74} \times \frac{10}{73} = 0.00326$
method— allow with replacement

M1

0.00326 ($0.00325 \sim 0.0033$) or $\frac{44}{13505}$ acf

A1

(ii) $\frac{45}{75} \times \frac{44}{74} \times \frac{43}{73} + \frac{30}{75} \times \frac{29}{74} \times \frac{28}{73} = 0.270$

M1

attempt at $P(\text{all female}) + P(\text{all male})$

M1

*method for $P(\text{all female})$ and $P(\text{all male})$ –
allow with replacement*

0.270 ($0.2695 \sim 0.271$) or $\frac{730}{2701}$ acf

A1

5

[13]

Q11.

- (a) $0.5 \times 0.2 = 0.1$ (or 10%)

M1 A1

2

- (b) $0.1 + (0.3 \times 0.4) + (0.2 \times 0.1) =$
for 'their' 0.1 + considering other two

M1

for 0.3×0.4 or 0.2×0.1 effort

M1

$$0.1 + 0.12 + 0.02 = 0.24 \text{ (or 24\%)}$$

A1

3

(c) $\frac{0.1}{0.24} = 0.417 \text{ (or 41.7\%)}$
for numerator

M1

for denominator ft

M1

A1

3

(d) $\frac{(0.5 \times 0.8)}{1 - 0.24} = 0.526 \text{ (or 52.6\%)}$
for numerator (B1 for $0.8 \times 0.5 = 0.4$)

M1

for denominator ft part(b)

M1

or $\frac{40}{76}$ or $\frac{10}{19}$

A1

3

[11]

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Q12.

(a) (i) $145/250 = 29/50 = 0.58 \text{ (or 58\%)}$

B1

(ii) $80/250 = 8/25 = 0.32 \text{ (or 32\%)}$

B1

(iii) $120/250 = 12/25 = 0.48 \text{ (or 48\%)}$

B1

(iv) $65/120 = 13/24 = 0.542 \text{ (or 54.2\%)}$

M1

for using 120

A1

5

(b) (i) $1 - (0.8)^4 = 0.590 = 5904/10000$
for 0.8^4

M1

(or 59.0%)

A1

2

(ii) $P(+ve \text{ response}) =$
 $(0.2 \times 0.9) + (0.8 \times 0.15)$

M1

$= 0.3$

A1

$P(\text{not suffering} \mid +ve \text{ response}) =$
 $(0.8 \times 0.15) / 0.3$

M1

$= 0.12 / 0.3 = 0.4 \text{ or } 2/5 \text{ (or 40\%)}$

A1

4

[11]

Q13.

(a) (i) $\frac{3}{28} = 0.107$

M1

1

(ii) $\frac{5}{28} = 0.179$

M1

1

(iii) $\frac{3}{13} = 0.231$

M1

$0.107 \text{ (0.1065 to 0.1075)}$

A1

2

0.231 (0.23 to 0.231)
 0.179 (0.178 to 0.179). *all acf*

(b) (i) $\frac{6}{28} \times \frac{5}{27} = 0.0397$
allow with replacement

M1

0.0397 (0.0396 to 0.04) *acf*

A1

2

(ii) $2 \times \frac{15}{28} \times \frac{13}{27} = 0.516$
allow with replacement

M1

0.516 (0.516 to 0.52) *acf*

A1

2

(c) (i) S,T
S, T cao

B1

1

(ii) $P(S) \neq P(S | R)$ ($0.179 \neq 0.231$)
Reason

M1

No

No - needs numerical support

A1

2

or $P(R).P(S) \neq P(R \cap S)$

$$\frac{13}{28} \cdot \frac{5}{28} = 0.0829 \neq \frac{3}{28} = 0.107$$

or $P(R) \neq P(R | S)$ $\left(\frac{13}{28} \neq \frac{3}{5} \right)$

[11]

Q14.

(a) (i) $\frac{87}{170}$ or 0.512 or 51.2%

B1

1

(ii) $\frac{18}{170}$ or $\frac{9}{85}$ or 0.106 or 10.6%

B1

1

(iii) $\frac{96}{170}$ or $\frac{43}{85}$ or 0.565 or 56.5%
For attempt at 18+46+23+9

M1

A1

2

(iv) $\frac{9}{83}$ or 0.108 or 10.8%
For numerator

M1

For denominator

M1

A1

3

(b) Resident occupies a two-bedroomed

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E1

apartment and replies excessive

Correct description

No negatives, description clear

If —given” used E1 only

E1

2

[9]

Q15.

(a) (i) $0.4 \times 50 + 0.1 \times 30 = 23$
M1 for 0.4×50 etc
M1 for total 23

M1 M1

$$\text{prob} = \frac{23}{80} \text{ or } 0.287(5)$$

A1 correct (Accept %)

A1

3

(ii) $\frac{20}{23}$ or 0.870

M1 for denomination
A1 correct

M1 A1

2

(b) (i) $0.10 + 0.20 - 0.25 = 0.05 \left(\frac{1}{20} \right)$

M1A1

2

(ii) $\frac{0.05}{0.20} = \frac{5}{20} = \frac{1}{4} = 0.25$

for denominator

M1 A1

2

[9]

Q16.

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(a) (i) $\frac{38}{150} = 0.253$

M1

(ii) $\frac{3}{79} = 0.0380$

0.253 (0.253 – 0.254) and
0.0389 (0.379 – 0.38) or acf

M1 A1

3

(b) $6 \times \frac{38}{150} \times \frac{84}{149} \times \frac{285}{148} = 0.162$

6

method - allow without replacement, or

omitted or incorrect 6
 Completely correct method
 0.162(0.162 – 0.163) or acf

B1 M1 m1 A1

4

(c) (i) $\frac{38}{150} \times 0.5 = 0.127$

M1 A1

2

(ii) $\frac{38}{150} \times 0.5 + \frac{84}{150} \times 0.25 + \frac{38}{150} \times 0.75 = 0.407$

Addition of 3 probabilities

Completely correct method A1 3

0.407 (0.406 – 0.407) or acf

M1 m1

(iii) $\left(1 - \frac{19}{150}\right)\left(1 - \frac{19}{149}\right) = 0.762$

Reasonable attempt

Wholly correct method - allow without replacement - allow at least 4

0.762(0.761 – 0.7625) or acf

M1 m1 A1

3

[15]

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Q17.

(a) $0.65 \times 0.46 = 0.299$

M1 A1

2

(b) $0.65 + 0.52 - 0.30 = 0.87$

For 0.65 + 0.52

M1 A1

2

(c) $1 - (0.52 + 0.38 - 0.25) = 0.35$

For 0.52 + 0.38 – 0.25

For sensible effort at 1–

M1M1A1

3

(d) $\frac{0.65 \times 0.46}{0.38} = 0.787$

For numerator

ft part (a), (not 0.65×0.38) M1 A1 3

For denominator

M1

[10]

Q18.

(a) (i) $P(S) = \frac{57}{174} = \frac{19}{58} = 0.328$

B1

1

(ii) $P(T) = \frac{18}{174} = \frac{3}{29} = 0.103$

B1

1

(iii) $P(R \cup T) = \frac{64}{174} = \frac{32}{87} = 0.368$
for attempt at $28 + 13 + 18 + 1 + 4$

M1 A1

2

(iv) $P(H \cap R') = \frac{16}{174} = \frac{8}{87} = 0.0920$

M1

for attempt at $1 + 4 + 4 + 7$

A1

2

(v) $P(S | T) = \frac{1}{18} = 0.0556$
for 1

M1

for 18

M1

A1

3

(iv) $P(R | H') = \frac{28}{127} = 0.220$
 for 28

M1

for 127

M1

sc 0.221

A1

3

(b) $P(R \cap H \cap T) = \frac{13}{174} = 0.0747$
 for 13

M1 A1

Male aged 55 – 74 years (with high blood pressure)
 who is undergoing treatment

Not “given”

B1

3

[15]

Q19.

(a) $0.6^3 = 0.216$

M1

0.216 cao acf

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A1

2

(b) $1 - 0.6^3 - 0.4^3$

Attempt at $1 - P(GGG) - P(LLL)$ oe

M1

Completely correct method

m1

$= 0.72$

0.72 cao acf

A1

3

(c)
$$\begin{array}{l} L | LGL \\ L | GLL \end{array} \quad \begin{array}{l} 0.4 \times 0.6 \times 0.4 \\ + 0.6 \times 0.4 \times 0.4 \end{array}$$

Reasonable attempt at enumerating possibilities

M1

Correct method

m1

$$= 0.192$$

0.192 cao

A1

3

(d) 0.4

M1A1

2

- (e)
- | | | |
|----------|----------------------|----|
| L GGG | 0.6^3 | |
| L LGGG | $+ 0.4 \times 0.6^3$ | |
| L GLGG | $+ 0.4 \times 0.6^3$ | m1 |
| L GGLG | $+ 0.4 \times 0.6^3$ | |

Reasonable attempt at enumerating possibilities

M1

Reasonable attempt at evaluating probability

m1

Completely correct method

m1

$$= 0.475$$

0.475 (0.475 to 0.4755)

A1

4

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[14]

Q20.

- (a) (i) $P(B \cup C) = P(B) + P(C) - P(B \cap C)$
 $= P(B) + P(C) - P(B) \times P(C)$

For attempt at formula

M1

For independent events

M1

sc M2 if $P(C) = 0.28 - 0.1 + 0.02$

Let $x = P(C)$

$$0.28 = 0.1 + x - (0.1x)$$

For attempt at $x = P(C)$

M1

$$\text{so } 0.18 = 0.9x \quad x = 0.2$$

*For showing $P(C) = 0.2$ or using 0.2
and demonstrating 0.28*

ag

A1

4

$$\begin{aligned} \text{(ii)} \quad P(C' \cup B') &= P(C') + P(B') - P(C' \cap B') \\ &= 0.8 + 0.9 - (0.8 \times 0.9) \end{aligned}$$

M1

For attempt at formula

A1

2

$$\text{(b) (i)} \quad 0.1 \times 0.7 \times 0.8 = 0.056$$

M1 A1

2

$$\text{(ii) (i)} \quad + (0.9 \times 0.7 \times 0.2)$$

M1

$$+ (0.9 \times 0.3 \times 0.8)$$

M1

$$= 0.056 + 0.126 + 0.216 = 0.398$$

A1

3

$$\text{(iii)} \quad \frac{0.216}{0.398} = 0.543$$

For 0.216 f.t.

M1f.t.

For 0.398 f.t.

M1f.t.

A1

3

[14]

Q21.

(a) (i) $0.4 \times 0.7 = 0.28$

Method

M1

2

0.28 cao

A1

(ii) $(1 - 0.4)(1 - 0.35) = 0.39$

Method (answer given)

M1

0.39 cao

A1

2

(iii) $1 - 0.28 - 0.39 = 0.33$

Method

M1

0.33 cao

A1

2

(b) (i) $0.4 \times 0.7 \times 0.55 = 0.154$

Method

M1

0.154 cao

A1

2

(ii) $0.6 \times 0.35 \times 0.25 = 0.0525$

Method

M1

0.0525 (0.052 – 0.053)

A1

2

[10]

Q22.

(a) $0.6 \times 0.55 = 0.33$ or $\frac{33}{100}$ or 33%

M1 A1

2

(b) $0.4 \times 0.65 = 0.26 \left(\text{or } \frac{26}{100} \text{ or } 26\% \right)$

M1 A1

2

(c) $(0.26) + (0.6 \times 0.45) = 0.53$

M1

$$\left(\text{or } \frac{53}{100} \text{ or } 53\% \right)$$

For 0.6×0.45

M1 ft.

ft their (b) + above

A1

3

(d) $\frac{0.26}{0.53} = 0.491 \left(\text{or } \frac{26}{35} \text{ or } 0.491 \right)$

For numerator

M1 ft.

For denominator

M1 ft.

A1

3

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[10]

Q23.

(a) (i) $0.6 \times 0.55 = 0.33 \text{ (or } 33\%)$

M1A1

2

(ii) $0.4 \times 0.65 = 0.26 \text{ (or } 26\%)$

M1A1

2

(iii) $0.26 + (0.6 \times 0.45) = 0.53 \text{ (or } 53\%)$

M1A1

2

(iv) $\frac{0.26}{0.53} = 0.491$ (or $\frac{26}{53}$ or 49.1%)

M1A1

2

(b) P (female and in hall) =

$(0.6 \times 0.55) \times \frac{1}{2}$ (Arts)

$+ (0.4 \times 0.35) \times \frac{1}{3}$ (Science)
For both

M1

= 0.165 + 0.046

For adding

M1

= 0.2116

$P(\text{Science} | \text{female in hall}) = \frac{0.046}{0.2116}$

M1

= 0.220 (or 22% or $\frac{140}{635}$)

A1

4

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[12]

Q24.

(a) (i) $\frac{1}{3}$ (0.33 or better)
Method

M1

(ii) 0.4
Method

M1

(iii) 0.4
Clearly incorrect method M0
A1 all answers acf

M1 A1 4

- (b) (i) W, X
W, X

B1 1

- (ii) V, W
V, W

B1

Not mutually exclusive – same member may be female and adult
Reason not mutually exclusive – can be obtained for V, X

E1

Not independent $P(V|W) = \frac{5}{24} \neq P(V)(\frac{1}{3})$
 $P(V|W) \neq P(V)$

E1 3

[8]

Q25.

- (a) $0.2 \times 0.3 \times 0.1 = 0.006$ (or 0.6%)
Product of “not going” intended

M1 A1 2

- (b) $0.8 \times 0.3 \times 0.1 = 0.024$ (or 2.4%)
Correct product intended

M1 A1 2

- (c) $0.024 + (0.2 \times 0.7 \times 0.1) + (0.2 \times 0.3 \times 0.9)$
Attempt at other “one only goes” probs

M1

$= 0.024 + 0.014 + 0.054$
Summation of three

m1

$= 0.092$ (or 9.2%)

A1 3

(d) $\frac{0.024}{0.092} = 0.261$ (or $\frac{6}{23}$ or 26.1%)
Numerator
Denominator
(M1, M1, A0 for 0.26)

M1 M1 A1

3

[10]

Q26.

(a) $0.7 \times 0.8 \times 0.9 = 0.504$
Allow small slip
Completely correct method
0.504 cao, acf

M1 m1 A1

3

(b) $0.7 \times 0.2 \times 0.35 = 0.049$
0.049 cao, acf

M1 A1

2

(c) $0.3 \times 0.6 \times 0.55 = 0.099$
0.099 cao, acf

M1 A1

2

(d) $0.7 \times 0.8 \times 0.9 + 0.7 \times 0.2 \times 0.65 + 0.3 \times 0.6 \times 0.55 + 0.3 \times 0.4 \times 0.4 = 0.742$
Sum of 4 probabilities, at least one correct

M1

Completely correct method – allow one small slip
0.1742 cao, acf

m1 A1

3

[10]

Q27.

(a) (i) $\frac{8}{273} = 0.0293$ (or 2.93%)

B1

1

(ii) $\frac{82}{273} = 0.300$ (or 30.0%)

B1

1

(iii) $\frac{35}{82} = 0.427$ (or 42.7%)
M1 for use of 82

M1 A1

2

(iv) $\frac{47}{273} = 0.172$ (or 17.2%)
M1 for attempt at 47

M1 A1

2

(v) $\frac{122}{244} = 0.5$ (or $\frac{1}{2}$ or 50%)
For numerator
M1 for denominator

M1 M1 A1

3

- (b) a candidate who is female when it is known that the candidate had 50 or fewer lessons before they took their driving test
Correct
Simple (no 'not' used throughout) and correct use of 'given'

B1 B1

2

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[11]

Q28.

(a) $0.8 \times 0.3 \times 0.4 = 0.096$

M1 A1

2

(b) $0.8 \times 0.7 \times 0.6 = 0.336$

M1 A1

2

(c) $0.096 + 0.336 + (0.2 \times 0.7 \times 0.4) = 0.488$
Amy hitting
addition

M1 M1 A1

3

(d) $P(B \mid \text{exactly one}) = \frac{0.096}{0.488}$
numerator
denominator
ft 'their' (a) and (c) if sensible

M1 M1 A1

3

[10]

Q29.

(a) $0.18 \times 0.8 = 0.144$ or $\frac{18}{125}$

M1 A1

2

(b) $(0.144) + (0.82 \times 0.1) = 0.226$
for 0.82×0.1
for addition

M1 M1

or $\frac{113}{500}$

A1

3

(c) $\frac{(0.82 \times 0.1)}{0.226} = 0.363$ or $\frac{41}{113}$
for numerator
for denominator can
ft if sensible

M1 M1 A1

3

[8]

Q30.

(a) (i) $\frac{52}{170} = \frac{26}{85}$

For attempt to total for C

M1 A1

2

(0.306 or 30.6%)

(ii) $\frac{8}{170} = \frac{4}{85}$

(0.0471 or 4.71%)

For attempt to sum relevant frequencies

$45 + 17 + 8 + 35$

B1

1

(iii) $\frac{105}{170} = \frac{21}{34}$

M1 A1

2

(0.618 or 61.8%)

(iv) $\frac{8}{70} = \frac{4}{35}$

For use of 70

For use of 8

M1 M1

(0.114 or 11.4%)

A1

3

- (b) The student is either aged 16 – 17 years or replied “too high” or both

For 16 – 17 years and “too high”

B1

For including “or both”

B1

2

[10]

Q31.

(a) (i) $\frac{17}{170}$ (0.1 or 10%)

B1

1

(ii) $\frac{95}{170} = \frac{19}{34}$ (0.559 or 55.9%)

For sum attempt $45 + 17 + 8 + 25$
 or use of $P(D \cup B)$ formula M1
 $(33 + 70 - 8)$

M1 A1
 2

(iii) $\frac{17}{70}$ (0.243 or 24.3%)
 For using 70

M1 A1
 2

(iv) $\frac{162}{170} = \frac{81}{85}$ (0.953 or 95.3%)
 For sum attempt $45 + 17 + 40 + 35 + 25$
 (or $170 - 8$)

M1 A1
 2

(b) The student is 16-17 years and replied 'too high'
 Correct
 As simple as possible, no double negatives

B1 B1
 2

[9]

Q32.

(a) (i) $\frac{85}{200} = 0.425$
 cao acf

B1

(ii) $\frac{164}{200} = 0.820$
 Method
 cao acf

M1 A1

(iii) $\frac{34}{45} = 0.756$
 0.756 (0.755 – 0.756)

M1 A1

5

(b) (i) $\frac{90}{200} = 0.450$
cao acf

M1 A1

(ii) $\frac{90 \times 89 \times 88}{(200 \times 199 \times 198)} = 0.0894$
Allow with replacement
Completely correct method
0.0894(0.0894 – 0.0895) acf

M1 m1 A1

(iii) $\frac{3 \times 90 \times 89 \times 65}{(200 \times 199 \times 198)} = 0.198$
For 3
Allow 6, allow with replacement
0.198 (0.1975 – 0.199) acf

B1 M1 A1

8

[13]